

Exploring preparation practices of nursing and midwifery educators for in situ simulation-based education: A qualitative study in a large healthcare service

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ABSTRACT

Aim: This study aimed to explore nursing and midwifery clinical educators' preparation practices related to *in situ* simulation-based education, at a tertiary health service in Australia.

Background: Simulation-based education is routinely used in healthcare education and training. A key mechanism to optimise simulation-based education is learner preparation. While diverse pre-simulation preparation approaches are described in the literature, these are predominantly focused on activities that are undertaken in either university and/or skills centre contexts. In contrast, the learner preparation practices for simulation-based education that is delivered *in situ* in healthcare facilities by clinical nurse and midwifery educators are underexplored.

Design: This study used an exploratory qualitative design.

Methods: Participants were recruited using purposive sampling from a potential study group of thirty. Twelve individuals from the nursing and midwifery education group met selection criteria and agreed to participate in the study. Each individual participated in a semi-structured interview. Interview data were then transcribed and analysed using qualitative descriptive methods.

Results: Data analysis resulted in the development of four themes related to the preparation of participants for *in situ* simulation-based education sessions. Each theme informs the choices of clinical nurse/midwifery educators' preparation practices: 1) responsivity to workplace and clinical priorities; 2) clinical educator objectives; 3) preparedness for learning and clinical practice; and 4) evolving educational expertise.

Conclusion: This study explored the preparation practices of nurse and midwifery clinical educators engaged in the delivery of *in situ* simulation-based education. Findings indicate a depth of experiences and a willingness to share practice insights, suggesting that *in situ* simulation-based education is an integral part of the in service repertoire and a key component of departmental education strategies, designed to support practice and hone skills required to deliver quality patient care.

1. Introduction

Simulation-based education (SBE) is a well-established teaching and learning strategy within healthcare education and academic settings, its application in healthcare workplaces requires further exploration (Bryant et al., 2020; Clapper, 2013; Leigh and Steuben, 2018; McDermott,

2016; Roh et al., 2018; Tyerman et al., 2016, 2019). SBE practices are informed by experiential learning theory, building on the foundational works of Lewin, Dewey and others (Kolb & Kolb, 2009) and constructivist philosophy which assumes that learners actively build links between their existing knowledge and new knowledge (Biggs, 1996). Beyond higher education, SBE is increasingly embedded in healthcare

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organisations (Davies et al., 2023) as an approach to supporting the continuing professional development of healthcare professionals and teams. By situating their simulation sessions insitu clinical nurse and midwifery educators are enabled to delivery SBE training within the constraints of established in-service education timeframes and in response to emergent clinical education needs.

SBE delivery in healthcare institutions has the advantage of being responsive to workplace and clinical priorities and, therefore, objectives can vary by context and location. Simulation providers deliver SBE sessions in a range of locations across healthcare workplaces to meet the unique demands of these institutions. SBE is a mainstay of courses such as Advanced Life Support (ALS) that are usually conducted in purpose-built skill centres, SBE sessions also happen in dedicated simulation spaces in healthcare departments such as the emergency department (ED) and intensive care unit (ICU) known as on-site training opportunities for clinical teams. Insitu simulation opportunities are ones that make use of actual clinical spaces where staff work, enabling opportunities to test real systems and latent safety threats in the environment. (Petrosoniak et al., 2017).

The existing literature on simulation delivery practices broadly focuses on scenario design, enactment and debriefing (Leigh and Steuben, 2018). SBE preparation, including both pre-simulation and pre-briefing activities, emerging as an important element of simulation, a recent systematic review reported that simulation participants benefited from pre-simulation activities in terms of satisfaction, knowledge and skill performance when these activities were tailored to their level of clinical and simulation experience (Tyerman et al., 2019). Additionally, there are well documented challenges associated with pre-simulation learner anxiety and difficulties engaging with 'fake' patients in inauthentic environments (Carey and Rossler, 2023). Despite simulation practitioners reporting essential preparation for SBE is important for learner success, there is limited guidance related to pre-simulation preparation activities in the workplace settings for post-licensure nurses/midwives (Rutherford-Hemming et al., 2019). Until recently there was no definitive agreement on the structure, format, or design of preparation activities (Leigh and Steuben, 2018; McDermott, 2016; Page-Cuttrara, 2015; Page-Cuttrara and Turk, 2017; Sharoff, 2015; Tyerman et al., 2016, 2019).

Recently the International Nursing Association for Clinical Simulation and Learning (INACSL) standards committee published a standard to offer guidance to educators from all disciplines as they develop their simulation activities (McDermott et al., 2021). While the INACSL standard recognised the need for participants to be prepared for simulation-based experiences, it focuses on provide broad advice rather than specific pre-simulation preparation activities. Given the emerging focus on pre-briefing and preparation for SBE, the opportunity to explore current preparation practices of nursing and midwifery clinician educators who were already delivering SBE as part of their education strategy, was identified.

2. Methods

This study used a descriptive, qualitative design to explore preparation practices of nursing and midwifery educators to gain rich insights into SBE preparation practices in one health service (Nestel et al., 2019). A purposive sampling method was used to recruit participants with the relevant deep experience in simulation-based education (Nestel, D. & Calhoun, A.W. 2019). Participants were recruited from the Gold Coast Hospital and Health Service (GCHHS) nursing and midwifery education teams, spanning diverse clinical areas including emergency, medicine, surgery, paediatrics and midwifery. Potential participants were identified on the basis that they: 1) were employed as a nurse educator, midwifery educator, clinical facilitator, or practice development nurse/midwife; 2) had been involved in delivering insitu SBE in their clinical areas over the 12–18 months prior to recruitment; and 3) were considered if they had previous employment as a clinical educator

during this timeframe.

Approximately 30 individuals who met inclusion criteria for the study were identified, with 12 individuals successfully recruited to the study. Recruitment of individuals who met the selection criteria for the study was done via an internal Microsoft Teams education group. Interested individuals were required to voluntarily respond with an expression of interest via email to the lead researcher (JS). Subsequently these individuals were provided with a participant information sheet, then asked to sign and return a study consent form to the lead researcher. From this recruitment process, twelve participants, all experienced, female clinical educators, were recruited to the study. It should be noted that only one male clinical educator met criteria for this study and did not submit an expression of interest to be recruited to the study.

The study was undertaken in the latter half of 2020, at GCHHS – a large tertiary health service incorporating a 750-bed tertiary-level facility, a secondary 365-bed facility, a full-time day hospital, a satellite minor injuries facility and several community hubs and is credentialed as part of the MAGNET © recognition program (About Us Gold Coast Health, 2020). The education and research division of GCHHS is home to a dedicated simulation service is responsible for collaborating and partnering with clinical educators from a range of health professions (e. g., nursing, midwifery, medicine, allied health) to deliver SBE across the organisation. At the time of our study, approximately 21 clinical areas were actively engaged in the delivery of insitu SBE (Myers, 2020), annualised data indicating estimating 11,000 hours of SBE delivery including a significant proportion delivered insitu (Scott, 2021)

2.1. Data collection

A pilot interview was conducted prior to the finalisation of the interview guide. Individual semi-structured interviews were conducted by SL and JS in 2020. Research memos were used throughout interviews to sign post concept of interest. Data saturation was deemed to have been achieved following the conclusion of the 12th interview, data saturation in this study was characterised by the repeated emergence of concepts (Nestel and Calhoun, 2019). The principal interviewer was known to participants, however there was no reporting relationship with participants, therefore reducing the power differential. Interviews were, on average, 45 minutes in duration and were guided by a semi-structured interview guide (Table 1).

Interviews were audio recorded and transcribed to facilitate data analysis. Transcription was facilitated using a digital recording application, transcribed to word processing software and manual corrections were made, where required.

Transcripts were analysed using a process of reflexive thematic analysis (Bearman, 2019). Familiarisation with the dataset occurred during the transcript review stage (as outlined above). Transcripts were then independently reviewed by JS, SL and VB. Each reviewer made initial notes to identify codes and initial themes to be considered by the research group. Once this phase was complete researchers, met in person and using an iterative collaborative discussion based approach compared initial analyses, discussed potential themes and negotiated discrepancies before reaching consensus on the final themes (Braun and Clarke, 2012; Clarke and Braun, 2017; Gromley, et al. 2019) (Table 2). Ethics approval was sought and obtained from the GCHHS Ethics Committee and Flinders University (HREC/2020/QGC/65186).

3. Results

In examining the SBE preparation practices of clinical educators, four themes were developed from the interview data. These are: responsiveness to workplace and clinical priorities; educators' objectives; preparedness for learning and clinical practice; and evolving educational expertise.

Table 1
Semi-structured interview questions.

1. Please tell me about your current role as a clinical educator (Prompts: How long you have worked in nurse education? How long you have worked simulation-based education?)
2. Tell me about the training opportunities you have had that have helped you to develop your role as a clinical educator (Prompt: What has been most helpful in assisting you in your current role?)
3. What topics are covered through simulation in your clinical area? (Prompt: How do you decide which topics to cover?)
4. How do you prepare staff to engage in an on-site or in-situ simulation exercise? (Prompts: What considerations do you make in preparing staff for simulation exercises? What types of pre-simulation activities do you offer, and how do you decide what types of activities to offer? How do you provide materials to learners?)
5. What challenges have you experienced in relation to preparing learners to engage in simulation exercises?
6. Is there anything else you would like to share that we have not already discussed?

Table 2
Theme development.

Theme	Sub-Themes
Responsivity to workplace and clinical priorities	Meeting skills requirements Meeting workplace needs Bi-directionality
Educator Objectives	Psychological safety (learner preparedness) Optimising care (clinician preparedness)
Preparedness for learning and clinical practice	Preparation for the content of simulation Preparation for the process of simulation
Evolving educator expertise	

3.1. Responsivity to workplace and clinical priorities

Participants acknowledged the value of SBE in facilitating team-based learning in the workplace. They described a willingness to plan SBE that incorporated both review of familiar skills and the introduction and learning of new knowledge and skills required for the department. In planning SBE, participants reported that they engaged in an extensive planning process well before session delivery. This included strategic scenario design and planning, together with careful selection and curation of preparation activities, purposefully designed to meet existing and emerging departmental education priorities. For example, one participant described the introduction of new equipment in the hospital as a basis for SBE activity:

So this year, we have really chosen things where the staff needed to develop their knowledge ... We had the introduction of new defib machines, so we had a focus in our sims on that ... So in the lead up to those, we would do, time permitting, in-services just on the [resuscitation] cart and get the defib machines, make sure people know how to turn on, where you're going to put the pads ... (08)

This suggests that educators leverage an ongoing, 'bi-directional' relationship between meeting departmental needs and designing, planning and delivering SBE that is responsive to these needs.

3.2. Educator objectives

Participants reflected on their intentions, as clinical educators, when developing SBE preparation materials. This intentionality focused on learner preparedness for insitu SBE and enhancing clinician capability for service delivery.

3.2.1. Psychological safety (learner preparedness)

Participants held the view that learners' engagement with SBE preparation activities optimises learning during the simulation and active participation in the debrief. A fundamental objective that participants cited in designing preparation activities was to facilitate psychological safety ahead of the simulation. This was critical as participants recognised the role of anxiety in shaping learners' SBE experiences:

I think a lot of what hinders their learning is anxiety. I spend time trying to reassure them, that this is about learning. We're all here to participate and share and grow from each other (10).

3.2.2. Optimising care (clinician preparedness)

A second key objective for clinical educators was an emphasis on optimising care, improving clinical capability and enhancing service delivery, aligned with Theme 1, responsivity to workplace and clinical priorities. This contributed to their overall motivation to be involved in SBE, as one participant explained:

I'm wanting to help people to improve on their practice and focus in on supporting people to become good practitioners (12).

In various departments, participants were able to identify tangible improvements in clinical service delivery as a direct result of targeted SBE activity, such as the following example referring to pandemic preparation activities.

There are things in their practice that we can see, over time, from doing simulations that ...have improved (07).

3.3. Preparedness for learning and clinical practice

Similar to the objectives described by participants, two key components underpin preparation activities. This included a focus on (1) preparation for the *content* of the simulation and (2) preparation for the *process* of the simulation.

3.3.1. Content preparation

Preparing learners for the content of the simulation involved the identification of learning objectives, emphasising the knowledge and skills that learners would develop from engaging in the simulation activity. Participants described curating relevant and appropriate pre-reading materials aligned to intended learning outcomes:

... it is relevant to the case. It could be a blog...or a clinical case study ... (02)

In addition, participants described multi-faceted approaches to ensure that learners had could access and had adequate time to review preparation materials ahead of the simulation activity. Approaches that were cited included email bulletins, advertisements on education board newsletters and verbal reminder announcements at team huddles, as one participant explained:

... I'll email information to the team, I put it on the education board that is by the tearoom and dependent on the specific scenario...We might do some clinical skill stations prior to the simulation. (06)

3.3.2. Process preparation

Participants highlighted the importance of incorporating materials related to the learning process into preparation materials for learners. Included in this process preparation was information pertaining to the practicalities of participating in a simulated learning scenario such as outlining participation expectations (e.g., interactions with a simulated

patient or manikin, tasks to be performed such as administration of a medication). Educators described the use of participant information sheets to provide this information to learners prior to the SBE session, as one participant explained:

... we send out ... a participant guide for whatever the topic is ... It outlines what the simulation structure is like. So, like, the pre-brief, the actual debrief and then documents like the policy or, you know, say, 'Go and look for where's your specific equipment. You'll need to know where this is ... familiarise yourself with the contents.' (07)

In this instance, learners are directed to existing resources in their department, thereby facilitating familiarity with policies and procedures and equipment in the clinical environment. In this way, process preparation not only prepares learners for the simulation activity but increases knowledge and use of resources in the workplace.

3.4. Evolving educational expertise

Clinical educators recognised the need to consider their own educational journey to become adept at delivering SBE. Ongoing professional development and learning was evident in the way that participants described the evolution of their approach to and choice of, SBE preparation activities. Statements from participants such as:

"my approach has changed as I've learned more" (01) and "this has changed a lot over the years as my experience has developed" (04)

Participants explained that their knowledge and skills as a clinical educator were developed from multiple influences. Adapting approaches over time was derived from "trial and error" concepts and influenced by experiences and feedback:

"it was trial and error...so I like to trial things and see what response is and go off that...then I will sort of change and adapt" (10)

Several participants cited support from workplace mentors as having a profound positive influence on their development as clinical educators and in SBE. Whilst some participants described the importance of combining this mentoring with ongoing, formal, professional development to support how they developed their educational approach. One participant explained:

... for me, when I initially started, I would be learning from other CFs [clinical facilitators] and educators about how they do it. Then when I went to formal training, that's when I developed my own way of doing things." (05)

4. Discussion

This study explored the views of clinical educators about preparation practices for insitu SBE sessions delivered to nursing and midwifery teams. Findings indicate a depth of experiences and a willingness to share practice insights, suggesting that insitu SBE is an integral part of the in service repertoire and a key component of departmental education strategies, designed to support practice and hone skills required to deliver quality patient care.

In describing their SBE preparation activities, participants reflected on their experiences and clearly articulated their reasoning and rationales underpinning their approaches to preparing teams for SBE sessions. Preparation practices were characterised by a critical awareness of workplace and clinical priorities coupled with a willingness to be responsive and adaptive to these needs. In supporting these professional development needs, participants expressed purposeful and deliberate strategies designed to ensure that learners' needs were met from both a clinician learner and practitioner perspective. Preparation activities were evident at the level of 'curriculum', where participants articulated their intended learning objectives, but also demonstrated in the purposeful approach to the curation of preparation materials. Continuous

evaluation and improvement at the level of clinical service delivery was also reflected in a strong commitment to continually develop their educational expertise and thus SBE and the experience for the learner. Fig. 1 offers a conceptual model developed from the themes identified in this study that highlights clinical educators' decision-making and process when preparation for insitu SBE.

This study builds on contemporary understanding of the pre-simulation phase of SBE for insitu workplace sessions (Chamberlain, 2017; Rutherford-Hemming et al., 2019; Sharoff, 2015; Tyerman et al., 2019). Clinical educator preparation practices reported in this study reflect current recommendations (McDermott et al., 2021; Watts et al., 2021) about simulation design. These include incorporating multiple modalities, developing learning objectives and supporting learners' engagement with required foundational knowledge to facilitate immersion in the simulation experience (Leigh and Steuben, 2018; Page-Cuttrara and Turk, 2017; Sharoff, 2015; Tyerman et al., 2016, 2019). Participants demonstrated an understanding of the unique learning goals of teams and used this knowledge to inform decision-making regarding preparation activities (e.g., using skills revision and targeted pre-reading coupled with participant information guides and appropriate timing to support learner preparedness (Diaz and Anderson, 2021, Tyerman et al., 2016).

Clinical educator participants consistently relayed their belief that using pre-simulation activities psychologically safe learning environment for insitu SBE could be created. This group embraced the idea that, 'psychological safety plays a vital role in helping people overcome barriers to learning and change in interpersonally challenging work environments' (Edmondson et al., 2016). Indeed, thoughtful pre-briefing, introduction, orientation, or other similarly designed practices have been identified as an essential precursor for simulation to create a 'safe container' for learning (Rudolph et al., 2014). Further to this 'safe container' concept, Purdy et al., (2022) developed a model to outline the bidirectional impact of psychological safety and simulation in healthcare environments. Our findings reiterate the importance of bidirectional communication and feedback loops. In our study participants were thoughtful about their choice of simulation preparation activities and responsive to feedback from their participants about the relevance of these materials. Intentional review and integration of participant feedback was highlighted as important to improving participant SBE experience and supporting psychological safety.

Evolving educator expertise was shaped by engagement and collaboration with mentors and interactions with experienced simulation educators contributed to clinical educator confidence and agency in decision-making about SBE preparation activities (See Table 3). Supportive networks, mentors and communities of practice have been shown to enable clinical educators to share expertise and learn from and with each other (Symon and Walker, 2023). Facilitating networking and mentorship opportunities coupled with intentionally designed simulation faculty development courses are a key means by which organisations can support capacity building in their education teams and ultimately, optimise healthcare outcomes. We suggest that as educators embrace co-creation, collaboration and continuous improvement, new ideas for preparatory activities to support insitu SBE sessions and learner engagement will emerge and be tested. Such co-creation facilitates access to feedback from trusted colleagues to adapt and redefine resources and activities before they are shared with learners and has incredible potential to mediate and strengthen SBE communities of practice.

5. Limitations

This study was conducted in one Australian health service and was limited to obtaining perspectives from nursing and midwifery clinical educators. Additionally, only female perspectives were obtained. Further research could examine the perspectives of other health profession educators with a focus on obtaining a gender balanced sample to examine whether our findings are emulated in other workplaces

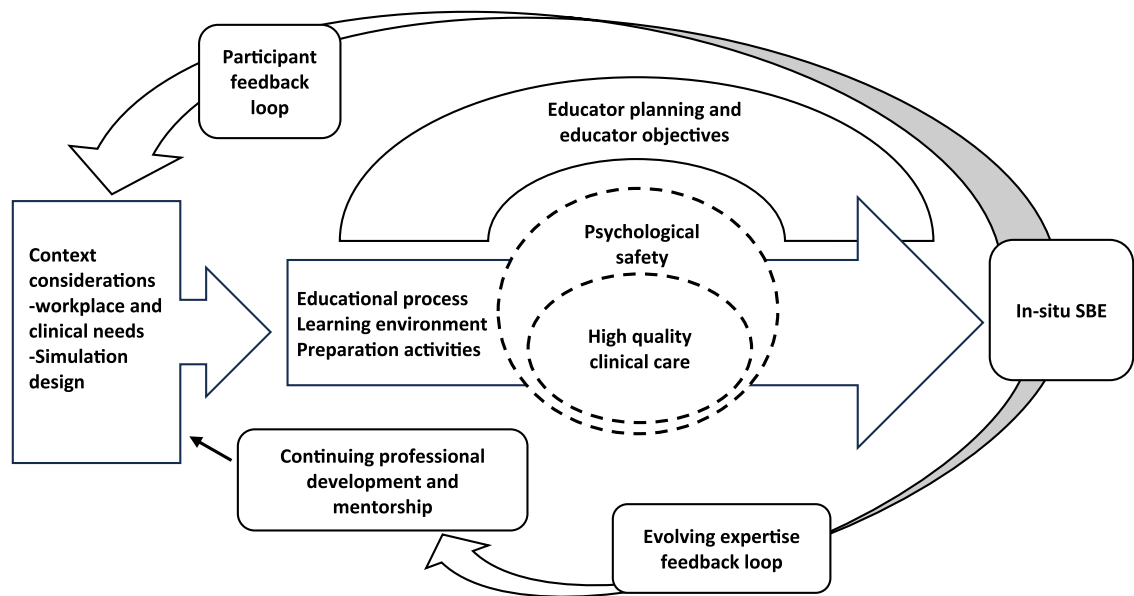


Fig. 1. Clinical educator processes in the preparation phase for in-situ simulation based-education (SBE).

Table 3

Key Practice Points.

Preparation
• Be mindful of workplace and clinical priorities • Consider content and process to prepare learners for SBE • Keep preparation materials short, sharp, and relevant • Use a combination of diverse approaches for preparation (e.g., participant information guides, micro skill teaching)
Educator Objectives
• Create a safe learning space ◦ Build into preparation activities ◦ Include pre-briefing as part of the simulation • Evaluate the impact of your SBE session on improving practice • Develop ways to build a learning culture within the team
Evolving Expertise
• Build connections with mentors, coaches, and a community of practice (local and online) • Continue to develop SBE skills through faculty development (local and online) and reflection on experiences.

involving insitu/on-site SBE. Additionally, this study focused on obtaining the views of those involved in the planning, design and delivery of SBE in different clinical contexts. Further research could triangulate these views with the perspectives of learners to explore the enablers and barriers to engagement with preparatory activities for insitu SBE.

6. Conclusion

This study explored nursing and midwifery clinical educator perspectives on the preparation phase of insitu SBE in a large health service. Our conceptual model highlights the multiple concepts and practices that inform insitu SBE preparation practices including emergent educational needs; the practice context; educators' objectives; and their evolving educational expertise. Preparation activities are strengthened with intentional co-design, in collaboration with clinicians and consumers, to have optimal relevance for participants. Designing feedback processes with multiple feedback loops, will support educators to reshape and improve the content and relevance of preparation activities for insitu SBE sessions, including the identification of barriers and enablers to activity completion. As educators continue to develop insitu SBE for workplace learning, further research is needed to inform how preparatory SBE activities add value to the overall SBE learning experience for the clinician learner.

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CRediT authorship contribution statement

Jane Schweitzer: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Svetlana King:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Victoria Brazil:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Samantha Lavender:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Letitia Del Fabbro:** Writing – review & editing, Writing – original draft, Visualization, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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